Project Proposal

Four Wheels

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Abstract

Four Wheels is a web-based online vehicle buying and selling platform designed to provide users with a secure and user-friendly environment for vehicle trading. The system allows sellers to list vehicles with complete details and images, while buyers can easily search, compare, and purchase vehicles based on city, price, and vehicle type. The platform focuses on providing a simple and user-friendly interface to enhance user experience. The proposed system aims to replace traditional and scattered vehicle trading methods with a centralized digital solution that ensures transparency, efficiency, and convenience.

Background of the Project

In the traditional vehicle buying and selling process, buyers and sellers rely on physical markets, dealers, or social media platforms, which often lack trust, verification, and complete information. This leads to time wastage, misinformation, and limited reach. With the growth of internet technology, online marketplaces have become popular, offering better accessibility and wider audience reach. The Four Wheels project is developed to address these issues by providing a modern digital platform where users can buy and sell vehicles easily, securely, and efficiently from anywhere.

Problem Statement

Vehicle buyers face difficulty in finding verified and complete vehicle information on a single platform. Sellers struggle to reach potential buyers due to limited exposure. Existing systems lack proper filtering, comparison, and visualization features. Manual communication methods increase the risk of fraud, misinformation, and delays in decision-making.

Solution Overview

- Online vehicle buying and selling platform
- Secure user registration and login system
- Advanced search by city, price, and vehicle type
- Direct buyer-seller communication
- Centralized and verified vehicle data

Objective of the project

General Objective:

To develop a modern, secure, and user-friendly online vehicle buying and selling platform using React.js for frontend, Node.js with Express.js for backend, and MongoDB for database management.

Specific Objectives:

- To develop a fast and responsive frontend using React.js.
- To build secure backend APIs using Node.js and Express.js.
- To store and manage vehicle and user data efficiently using MongoDB.
- To enable users to buy and sell vehicles online easily.
- To provide advanced search and filtering features.
- To enhance transparency and trust between buyers and sellers

Scope of the project

The scope of the Four Wheels project includes vehicle listing, searching, filtering, and secure communication between buyers and sellers. The system supports multiple vehicle categories and integrates a database-driven backend for efficient data management. Future enhancements may include online payments and mobile application support.

Methodology

The development of the Four Wheels system follows the Waterfall Methodology along with modern MERN stack technologies. The system is developed in phases including requirement analysis, system design, frontend development using React.js, backend development using Node.js and Express.js, database integration with MongoDB, testing, and deployment. This structured approach ensures smooth integration between frontend, backend, and database.

Reason for Chosen Methodology

The Waterfall Methodology was selected because project requirements are clearly defined and the system needs proper documentation at each stage. Using React.js, Node.js, Express.js, and MongoDB allows efficient development, scalability, and better performance. This combination makes the system reliable, easy to maintain, and suitable for a final year project.

- **Requirement Analysis and Definition**
- **System Software Design**
- **Implementation and Unit Testing**
- **Integration and System Testing**
- **Operation and Maintenance**

"Waterfall methodology" is phase developing methodology. Each phase is documented briefly and sends to the user for approval, if it is approved then the next phase is started. If the first phase is not approved by client or it does not according to the requirement of the client it is re-documented.

Because my project requires the same situation so "Waterfall Methodology" is best to use.

Development Tools

Activities	Tools/Programs
Frontend development	HTML, CSS, JavaScript
Backend development	Node.js + Express.js
Database	MongoDB
Platform	Windows
Browsers	Google chrome
Documentation	MS Word
Diagram	Draw.io

Table1

Expected Outcomes

- Easy and secure vehicle buying and selling
- Improved accessibility for buyers and sellers
- Efficient vehicle search and comparison
- Enhanced user experience through a fast and responsive interface
- Reduced fraud and misinformation

Conclusion

In conclusion, the Four Wheels project aims to modernize the vehicle trading process by providing a reliable and user-friendly online platform. By replacing traditional methods with a digital solution, the system enhances efficiency, transparency, and convenience for both buyers and sellers. The project is expected to make vehicle transactions simpler, faster, and more secure.